

Minimal prerequisites for Corollary 3.4 of Gurevich–Lapid*

This note contains exactly the definitions needed to *state* the single RSK step K (§2.2.2), the single Mœglin–Waldspurger step MW (§3.1), and Corollary 3.4. Tableaux, the recursive map RSK , and §§2.2–2.2.1 of the paper are not needed.

1. Segments and multisegments

[§2.1, p. 648]

Definition. A *segment* is a subset of $\mathbb{Z}$ of the form $\Delta = [a, b] = \{n \in \mathbb{Z} : a \leq n \leq b\}$ with $a \leq b$. Write $b(\Delta) = a$, $e(\Delta) = b$, and

$$\overleftarrow{\Delta} = [a - 1, b - 1], \quad {}^-\Delta = [a + 1, b] \quad (\text{so } {}^-\Delta = \emptyset \text{ if } a = b).$$

Seg denotes the set of all segments. For $\Delta_1, \Delta_2 \in \text{Seg}$,

$$\Delta_1 \prec \Delta_2 \iff b(\Delta_1) < b(\Delta_2) \text{ and } e(\Delta_1) < e(\Delta_2)$$

(a strict partial order). Also $\Delta_2 \subseteq \Delta_1$ iff $b(\Delta_1) \leq b(\Delta_2)$ and $e(\Delta_2) \leq e(\Delta_1)$.

A *multisegment* is a multiset of segments, written as a formal finite sum $\mathbf{m} = \sum_{i \in I} \Delta_i$. The set of multisegments is $\mathcal{M} = \mathbb{N}(\text{Seg})$. For $\mathbf{m} \neq 0$ set $\min \mathbf{m} = \min_{i \in I} b(\Delta_i)$. If $\mathbf{n} \leq \mathbf{m}$ (as multisets) we call $\mathbf{n}$ a *sub-multisegment* of $\mathbf{m}$.

A *ladder* is a nonzero multisegment of the form $\mathbf{l} = \sum_{i=0}^k \Delta_i$ with $\Delta_i \prec \Delta_{i-1}$ for $i = 1, \dots, k$ (its *standard form*).

2. The single RSK step K

[§2.2.2, pp. 649–650]

Fix $0 \neq \mathbf{m} = \sum_{i \in I} \Delta_i \in \mathcal{M}$.

Definition (Depth [p. 649]). The *depth* $d = d_{\mathbf{m}} : I \rightarrow \mathbb{Z}_{\geq 0}$ is

$$d(i) = \max\{j : \exists i_0 = i, i_1, \dots, i_j \in I \text{ with } \Delta_{i_r} \prec \Delta_{i_{r+1}}, r = 0, \dots, j-1\},$$

and $d(\mathbf{m}) = \max_{i \in I} d(i)$ is the depth of $\mathbf{m}$.

Fact ([p. 650, sentence above Lemma 2.1 — **do not skip**]). If $d(i) = d(j)$ then $\Delta_i = \Delta_j$ or $\Delta_i \subsetneq \Delta_j$ or $\Delta_j \subsetneq \Delta_i$. (Immediate from the definitions of $\prec$ and d ; it is what makes the enumerations below exist.)

Definition (The map K [p. 650]). For each $k = 0, \dots, d(\mathbf{m})$ choose an *admissible enumeration* $\{i_1^k, \dots, i_l^k\}$ of the fiber $d^{-1}(k)$, i.e. one with $\Delta_{i_1^k} \supseteq \dots \supseteq \Delta_{i_l^k}$, and set $j_k := i_l^k$. Let $i \mapsto i^\vee$ be the

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permutation of I whose cycle decomposition is $\{(i_1^k, \dots, i_l^k)\}_{k=0, \dots, d(\mathbf{m})}$. Define

$$\Delta'_i = [\mathbf{b}(\Delta_i), \mathbf{e}(\Delta_{i^\vee})] \in \text{Seg}, \quad i \in I, \quad (2.1a)$$

$$I' = \{j_0, \dots, j_{d(\mathbf{m})}\}, \quad I'' = I \setminus I', \quad (2.1b)$$

$$\mathbf{l}(\mathbf{m}) = \sum_{i \in I'} \Delta'_i, \quad \mathbf{m}' = \sum_{i \in I''} \Delta'_i \in \mathcal{M}. \quad (2.1c)$$

For every $i \in I'$,

$$\mathbf{b}(\Delta'_i) = \max_{j \in I: d(j)=d(i)} \mathbf{b}(\Delta_j), \quad \mathbf{e}(\Delta'_i) = \max_{j \in I: d(j)=d(i)} \mathbf{e}(\Delta_j). \quad (2.2)$$

The single RSK step is

$$K : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{M} \times \mathcal{M}, \quad K(\mathbf{m}) = (\mathbf{l}(\mathbf{m}), \mathbf{m}').$$

$\mathbf{l}(\mathbf{m})$ is the *highest ladder* and $\mathbf{m}'$ the *derived multisegment* of $\mathbf{m}$.

Remark.

- (i) The paper asserts (p. 650–651) that $K(\mathbf{m})$ does not depend on the choice of admissible enumerations. For our purposes one may instead *fix* a canonical enumeration (break ties between equal segments deterministically), so K is well-defined by construction; otherwise well-definedness is a small private lemma.
- (ii) By Lemma 2.1(1) of the paper (p. 650), $\mathbf{l}(\mathbf{m})$ is in fact a ladder. **This is not needed** to state Corollary 3.4; we take the codomain of K to be $\mathcal{M} \times \mathcal{M}$. If the ladder property is wanted, prove it as a private lemma from (2.2) and: for every $i \in I$ and $0 \leq k < d(i)$ there is $i' \in I$ with $\Delta_i \prec \Delta_{i'}$ and $d(i') = k$ (Lemma 2.1(1)).

3. The single MW step

[§3.1, pp. 654–655]

(This is the *mirror image* of the procedure of Mœglin–Waldspurger, obtained by inverting all inequalities; single step only — the full involution $\mathbf{m} \mapsto \mathbf{m}^\#$ is not needed.)

Definition. Let $0 \neq \mathbf{m} = \sum_{i \in I} \Delta_i$. Choose $i_1 \in I$ with

$$\mathbf{b}(\Delta_{i_1}) = \min \mathbf{m} \text{ and } \mathbf{e}(\Delta_{i_1}) \text{ minimal with this property.} \quad (3.1a)$$

Recursively: given i_j , if there is $i \in I$ with $\Delta_{i_j} \prec \Delta_i$ and $\mathbf{b}(\Delta_i) = \mathbf{b}(\Delta_{i_j}) + 1$, choose i_{j+1} among such i with $\mathbf{e}(\Delta_{i_{j+1}})$ minimal (3.1b)

and stop at the first k for which no such i exists. (3.1c)

Set

$$\Delta^\circ(\mathbf{m}) := [\min \mathbf{m}, \min \mathbf{m} + k - 1] \in \text{Seg}.$$

$I^* = \{i_1, \dots, i_k\}$ is a *set of leading indices* (not unique, but the multiset $\{\Delta_{i_1}, \dots, \Delta_{i_k}\}$ depends only on $\mathbf{m}$). Define

$$\Delta_i^* = \begin{cases} -\Delta_i & i \in I^*, \\ \Delta_i & \text{otherwise,} \end{cases} \quad \mathbf{m}^\dagger := \sum_{i \in I} \Delta_i^* \in \mathcal{M} \quad (\text{discarding empty summands}). \quad (3.2)$$

The single MW step is

$$\text{MW} : \mathcal{M} \setminus \{0\} \longrightarrow \mathcal{M} \times \text{Seg}, \quad \text{MW}(\mathbf{m}) = (\mathbf{m}^\dagger, \Delta^\circ(\mathbf{m})).$$

4. The statement

[Corollary 3.4, p. 660]

Two small facts are needed for the statement to be well-formed (both to be proved as private lemmas):

Lemma 1 (Well-formedness). *Let $0 \neq \mathbf{m} \in \mathcal{M}$ with $\min \mathbf{m} < \min \mathbf{l}(\mathbf{m})$. Then $\mathbf{m}' \neq 0$ and $\mathbf{m}^\dagger \neq 0$, so that $\text{MW}(\mathbf{m}')$ and $K(\mathbf{m}^\dagger)$ are defined.*

Proof sketch. If $\mathbf{m}' = 0$ then $\mathbf{m} = \mathbf{l}(\mathbf{m})$ up to the identification (2.1), forcing $\min \mathbf{l}(\mathbf{m}) = \min \mathbf{m}$. If $\mathbf{m}^\dagger = 0$ then every $\Delta_i = [\min \mathbf{m}, \min \mathbf{m}]$, and again $\min \mathbf{l}(\mathbf{m}) = \min \mathbf{m}$. (Cf. non-degeneracy, established in Proposition 3.3(1) of the paper under this hypothesis.) $\square$

Corollary 3.4 ([p. 660]). For any $0 \neq \mathbf{m} \in \mathcal{M}$ with $\min \mathbf{m} < \min \mathbf{l}(\mathbf{m})$ we have

$$(K \times \text{Id})(\text{MW}(\mathbf{m})) = (\text{Id} \times \text{MW})(K(\mathbf{m})).$$

In other words,

$$\mathbf{l}(\mathbf{m}) = \mathbf{l}(\mathbf{m}^\dagger), \quad \Delta^\circ(\mathbf{m}) = \Delta^\circ(\mathbf{m}'), \quad (\mathbf{m}^\dagger)' = (\mathbf{m}')^\dagger.$$

Source map

Segments, $\prec$, $\overleftarrow{\Delta}$, ${}^-\Delta$, multisegments, ladders, $\min \mathbf{m}$	§2.1, p. 648
Depth function $d_{\mathbf{m}}$	§2.2.2, p. 649 (bottom)
Comparability of equal-depth segments	p. 650, sentence above Lemma 2.1
Admissible enumerations, $i^\vee$, (2.1a)–(2.1c), (2.2), the map K	p. 650
“ $\mathbf{l}(\mathbf{m})$ is a ladder” (optional)	Lemma 2.1(1) + p. 650
$\Delta^\circ(\mathbf{m})$, leading indices, $\mathbf{m}^\dagger$, MW	§3.1, pp. 654–655
Corollary 3.4	p. 660 (top)